

Logging in TypeScript

Node.js & Express with Winston + Morgan

Why Logging Matters

"Your logs are your eyes in production"

- **Debugging** - understand what went wrong without a debugger
- **Monitoring** - track application health and performance
- **Auditing** - record important business events
- **Security** - detect and investigate incidents

The Problem with `console.log`

```
// ❌ Don't do this  
console.log("User logged in", userId);  
console.log("DB error", error);  
console.log("Request received");
```

Problems:

- No severity levels – everything looks the same
- No timestamps
- No file output – logs lost on server restart
- No environment-aware filtering

Log Levels

```
// Severity order (lowest number = highest severity)
const levels = {
  error: 0, // Something failed – needs immediate attention
  warn: 1, // Unexpected but handled – degraded behaviour
  info: 2, // Important business events – user login, order placed
  http: 3, // Incoming HTTP requests (Morgan uses this)
  debug: 4, // Detailed diagnostics for development
}
```

Rule of thumb: Production → warn , Development → debug

What TO Log

Application Events

- Service startup / shutdown
- User authentication (login, logout, failed attempts)
- Important state changes
- Background job completions

Technical Events

- Unhandled errors with stack traces
- External API calls (URL, status, duration)
- Database query failures
- Performance warnings (slow queries)

What NOT to Log

Never Log Sensitive Data

- Passwords or password hashes
- API keys / secret tokens
- Full credit card numbers
- Social security numbers

Avoid Logging

- Full request/response bodies (may contain PII)
- Session tokens or JWTs
- Personal identifiable information (PII)
- Medical or financial records

Logging Best Practices

- **Use structured logging** – log as JSON in production for easy parsing
- **Add context** – include user ID, request ID, correlation IDs
- **Be consistent** – same format, same level conventions throughout
- **Don't over-log** – avoid logging in tight loops or hot paths
- **Rotate logs** – prevent disk exhaustion with log rotation
- **Secure your logs** – treat log files as sensitive data

Setting Up Winston

```
npm install winston
```

Winston provides:

- Custom log levels & colours
- Multiple transports (console, file, external services)
- Flexible formatting (JSON, plaintext, timestamps)
- Environment-aware log level filtering

src/lib/logger.ts – Levels & Colours

```
import winston from 'winston'
```

```
const levels = {  
  error: 0,  
  warn: 1,  
  info: 2,  
  http: 3,  
  debug: 4,  
}
```

```
const colors = {  
  error: 'red',  
  warn: 'yellow',  
  info: 'green',  
  http: 'magenta',  
  debug: 'white',  
}
```

```
winston.addColors(colors)
```

src/lib/logger.ts – Dynamic Level by Environment

```
// Show all logs in development; only warn+ in production
const level = () => {
  const env = process.env.NODE_ENV || 'development'
  return env === 'development' ? 'debug' : 'warn'
}
```

Environment	Min Level	What You See
development	debug	Everything
production	warn	Warnings & errors only

src/lib/logger.ts – Format & Transports

```
const format = winston.format.combine(  
  winston.format.timestamp({ format: 'YYYY-MM-DD HH:mm:ss:ms' }),  
  winston.format.colorize({ all: true }),  
  winston.format.printf(  
    (info) => `${info.timestamp} ${info.level}: ${info.message}`  
  ),  
)  
  
const transports = [  
  new winston.transports.Console(),  
  new winston.transports.File({ filename: 'logs/error.log', level: 'error' }),  
  new winston.transports.File({ filename: 'logs/all.log' }),  
]
```

src/lib/logger.ts – Export the Logger

```
const Logger = winston.createLogger({  
  level: level(),  
  levels,  
  format,  
  transports,  
})  
  
export default Logger
```

Usage anywhere in your app:

```
import Logger from './lib/logger'  
  
Logger.error('Database connection failed', { host: dbHost })  
Logger.warn('Deprecated endpoint called')  
Logger.info('User logged in', { userId })  
Logger.debug('Computed result', { value })
```

Example log output (development)

```
2024-01-15 10:23:45:123 error: This is an error log
2024-01-15 10:23:45:124 warn:  This is a warn log
2024-01-15 10:23:45:125 info:  This is an info log
2024-01-15 10:23:45:126 http:  GET /users 200 - 12 ms
2024-01-15 10:23:45:127 debug: This is a debug log
```

- Each line is coloured by severity
- All logs written to `logs/all.log`
- Only errors written to `logs/error.log`

HTTP Request Logging with Morgan

```
npm install morgan @types/morgan
```

Morgan is an Express middleware that:

- **Automatically logs every HTTP request**
- Integrates with Winston via a custom stream
- Can be skipped in production

src/config/morganMiddleware.ts

```
import morgan, { StreamOptions } from 'morgan'
import Logger from '../lib/logger'

// Route Morgan output through Winston's http level
const stream: StreamOptions = {
  write: (message) => Logger.http(message),
}

// Only log requests in development
const skip = () => {
  const env = process.env.NODE_ENV || 'development'
  return env !== 'development'
}

const morganMiddleware = morgan(
  ':method :url :status :res[content-length] - :response-time ms',
  { stream, skip }
)

export default morganMiddleware
```

Wiring Morgan into Express

```
import express from 'express'
import Logger from '../lib/logger'
import morganMiddleware from '../config/morganMiddleware'

const app = express()
const PORT = process.env.PORT || 3000

// Register Morgan before your routes
app.use(morganMiddleware)

app.get('/health', (_, res) => {
  Logger.info('Health check called')
  res.json({ status: 'ok' })
})

app.listen(PORT, () => {
  Logger.info(`Server running on http://localhost:${PORT}`)
})
```


Morgan Token Format

```
:method :url :status :res[content-length] - :response-time ms
```

Produces logs like:

```
GET /api/users      200  342 - 23 ms
POST /api/users      201  128 - 45 ms
GET /api/users/99    404   48 - 8 ms
DELETE /api/users/1  403   60 - 5 ms
```

Each request is automatically logged – no manual `Logger.http()` calls needed.

Project Structure

```
src/  
├── lib/  
│   └── logger.ts           # Winston logger configuration  
├── config/  
│   └── morganMiddleware.ts # Morgan HTTP request logger  
├── routes/  
│   └── userRoutes.ts  
├── controllers/  
│   └── userController.ts  
└── index.ts                # Register morganMiddleware here  
logs/  
├── all.log                 # All log levels  
└── error.log               # Errors only
```

Security Considerations

```
// ❌ Never log sensitive data
Logger.info(`User login attempt`, {
  email,
  password, // NEVER log passwords
  token,    // NEVER log tokens
})

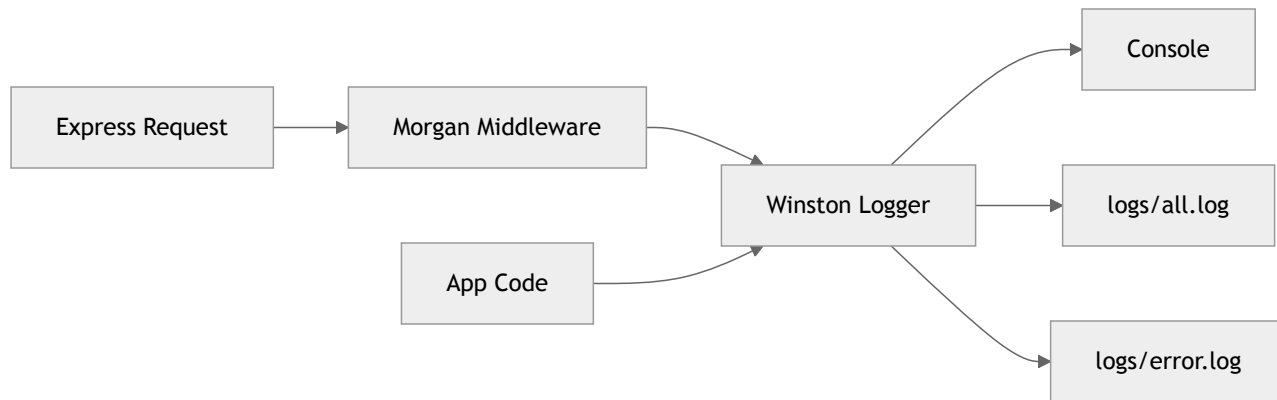
// ✅ Log only what you need
Logger.info('User login attempt', {
  email,
  ip: req.ip,
  userAgent: req.headers['user-agent'],
})
```

- Add `logs/` to `.gitignore` – never commit log files
- Apply file permissions: `chmod 640 logs/*.log`
- Consider a log management service (Datadog, Splunk) for production

Environment-Specific Configuration

Setting	Development	Production
Log Level	debug	warn
Console output	✓ Coloured	✓ JSON
File output	✓	✓
HTTP request logs	✓ All	✗ Skipped
Stack traces	✓ Full	✓ Full
PII in logs	✗ Never	✗ Never

Summary



- **Winston** – structured, level-aware application logging
- **Morgan** – automatic HTTP request/response logging
- Together they give you complete observability into your Express app

Key Takeaways

- Replace all `console.log` with Winston log levels
- Use `info` for business events, `error` for failures, `debug` for diagnostics
- Morgan handles HTTP request logging automatically – don't do it manually
- **Never log passwords, tokens, or PII**
- Adjust log verbosity by environment via `NODE_ENV`
- Add `logs/` to `.gitignore`

Further Reading

- Winston documentation
- Morgan documentation
- Article: Better logs for Express with Winston & Morgan
- OWASP: Security Logging and Monitoring Failures

Happy Logging! 🪵